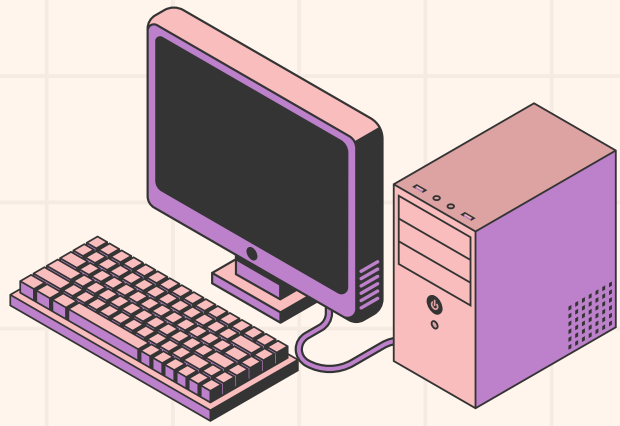


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INFECTIOUS DISEASE TRANSMISSION AND REPRODUCTION NUMBER

Modeling team - HCDC

NỘI DUNG TRÌNH BÀY

1

**Timeline of
infection**

2

**Infectious
measurements**

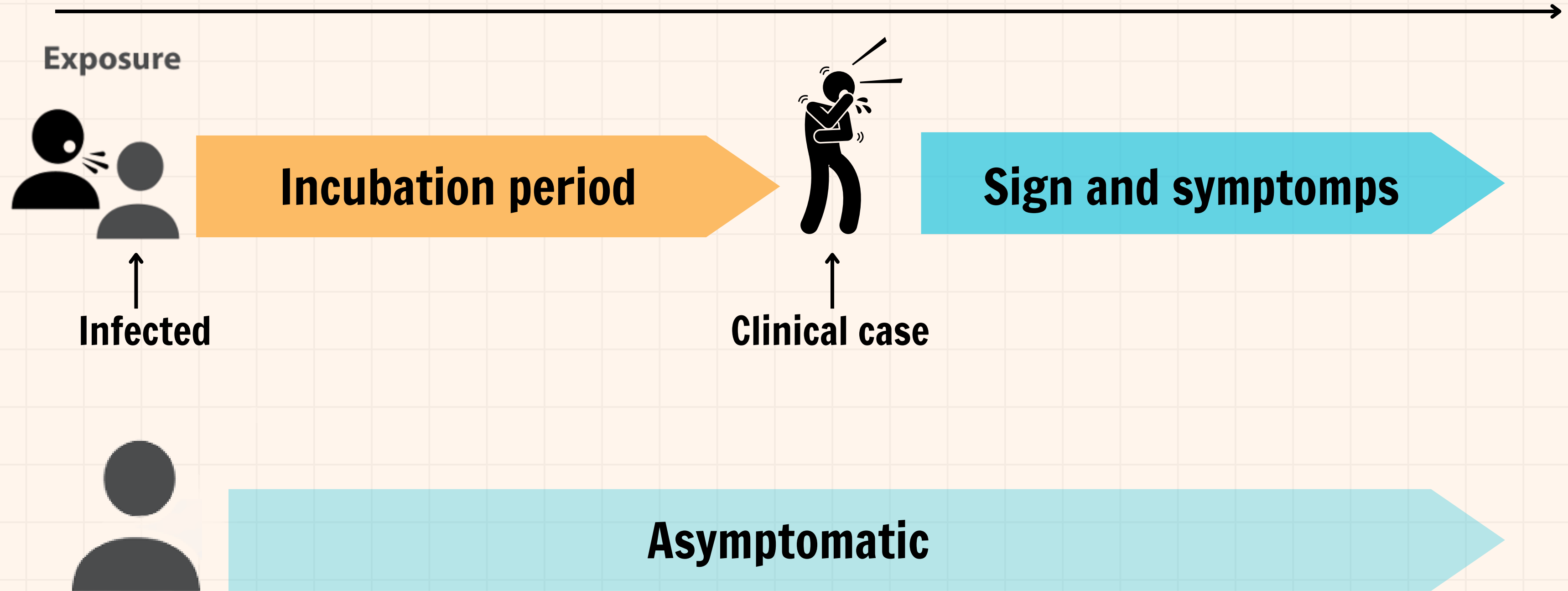
3

**Infectious
relationship**

4

**Reproduction
number**

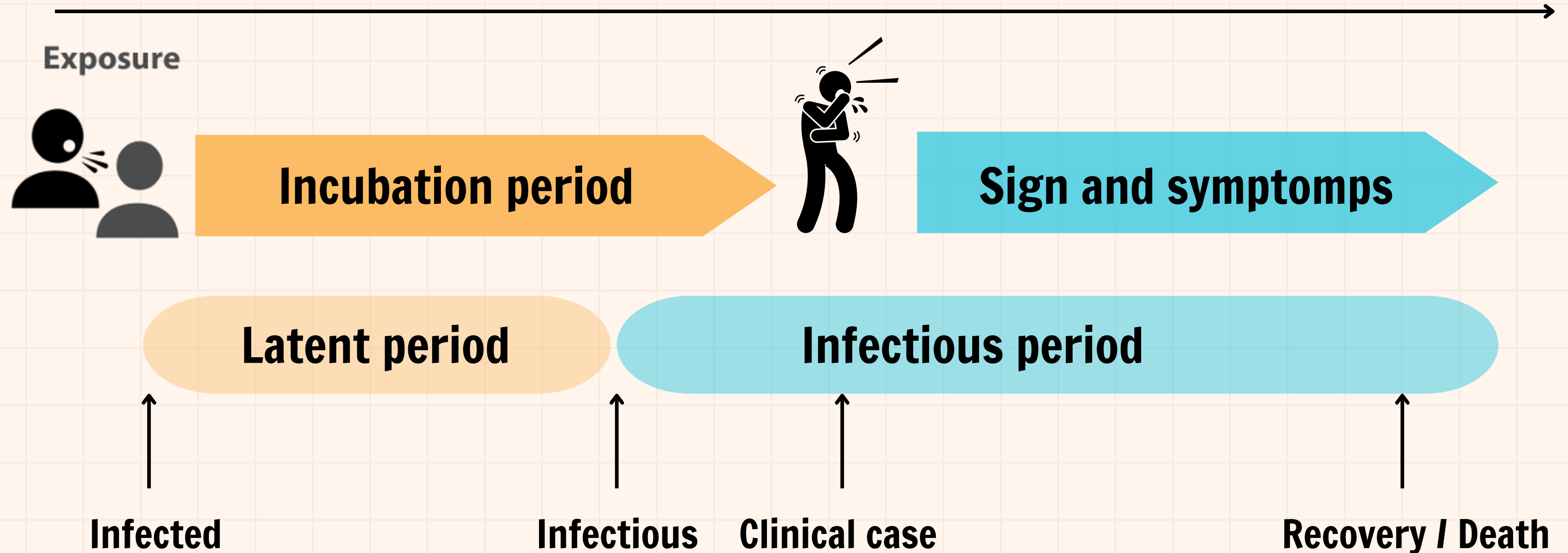
TIMELINE OF INFECTION



SWAPER

SWAPER

TIMELINE OF INFECTION



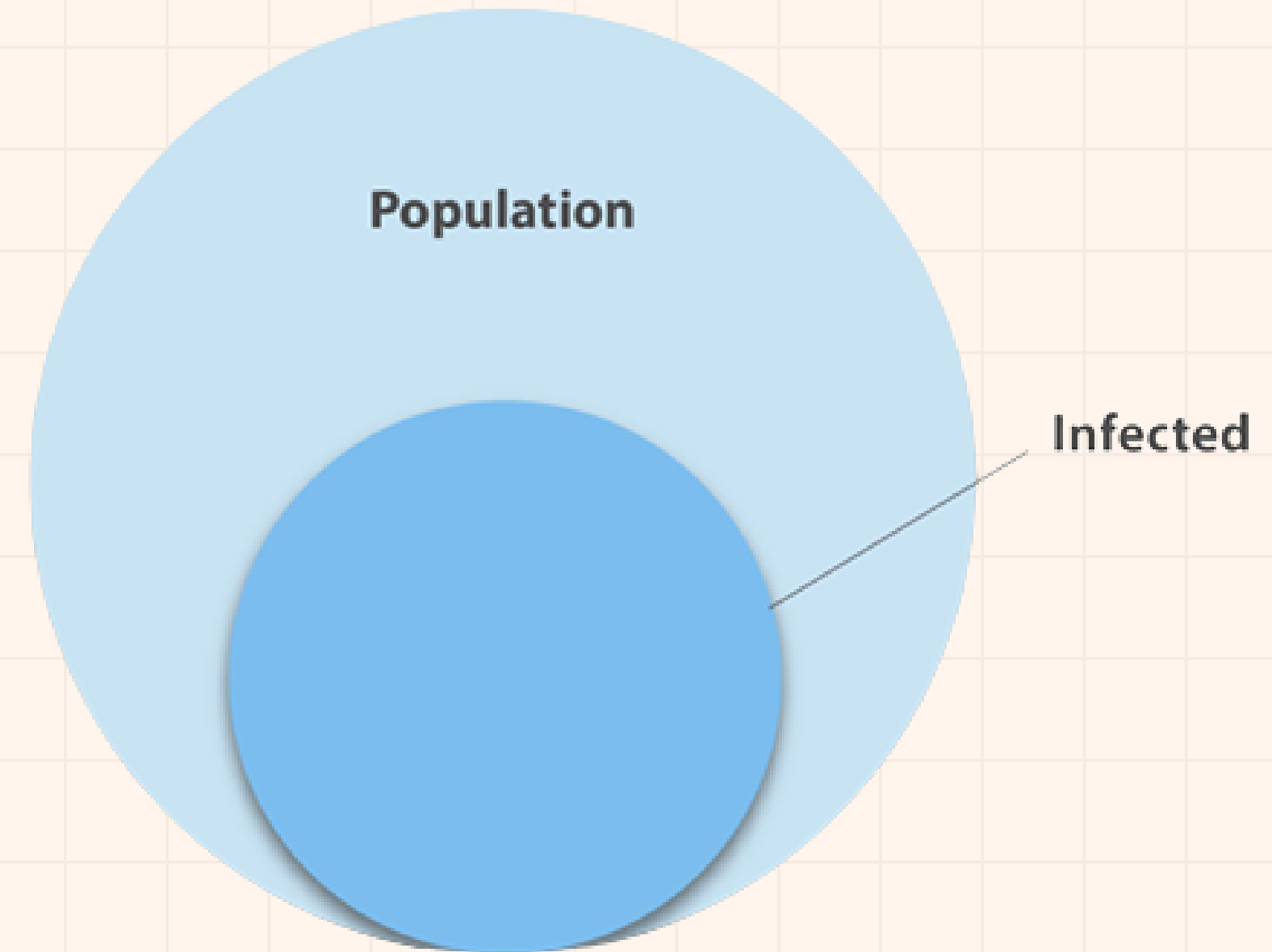
INFECTIOUS MEASUREMENTS

Prevalence rate

Number of infected
Population

Incidence

Number of new infected
Susceptibles



INFECTIOUS MEASUREMENTS

Pathogenicity

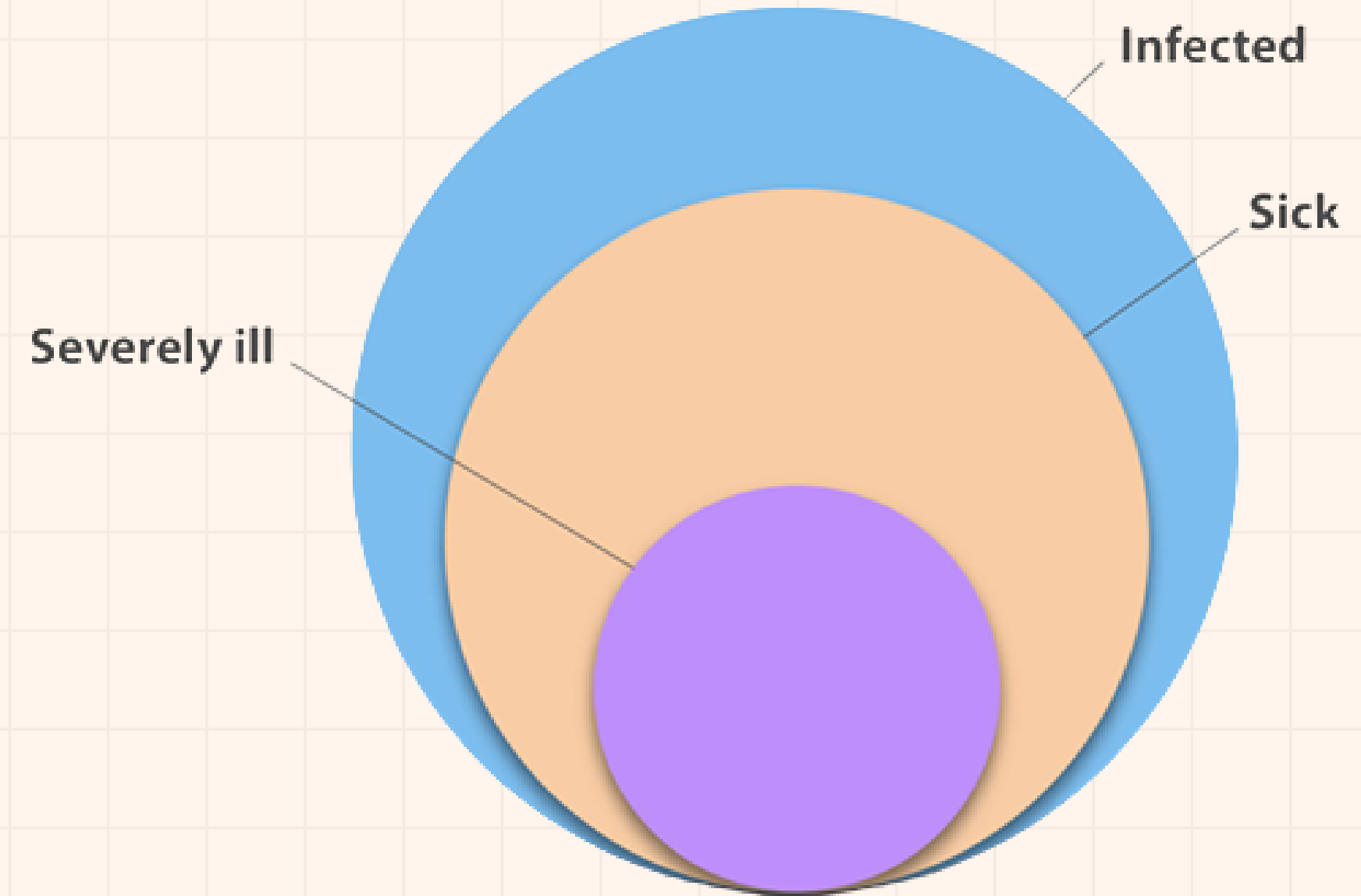
$$\frac{\text{Number of sick}}{\text{Number of infected}}$$

Asymptomatic

$$\frac{\text{Number of asymptomatic}}{\text{Number of infected}}$$

Virulence

$$\frac{\text{Number of severely ill}}{\text{Number of sick}}$$



INFECTIOUS MEASUREMENTS

Case fatality ratio

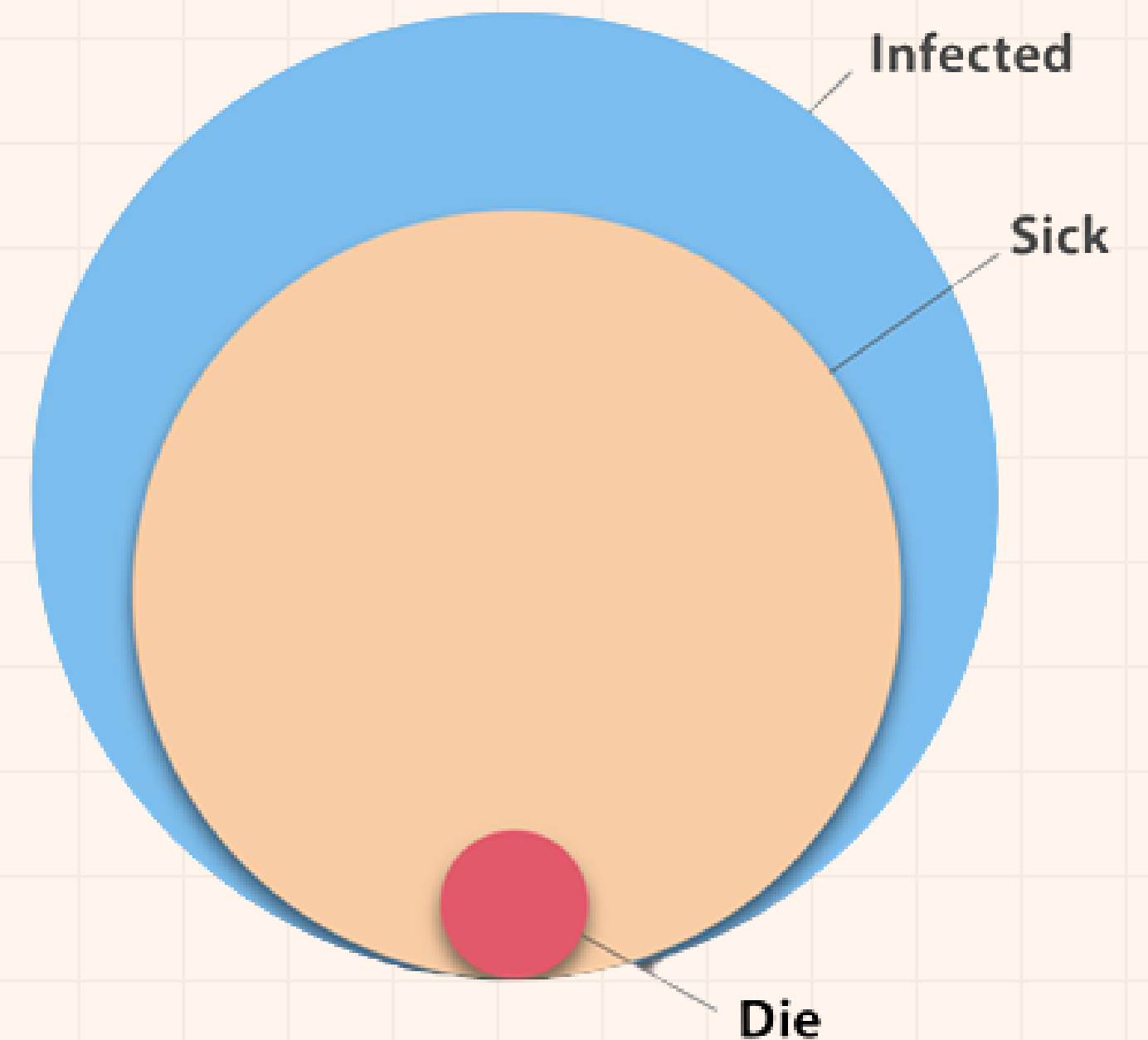
$$\frac{\text{Number who died}}{\text{Number of sick}}$$

Case fatality risk

$$\frac{\text{Number who died}}{\text{Number of sick}}$$

Case fatality rate

$$\frac{\text{Number who died}}{\text{Number of sick}}$$



INFECTIOUS MEASUREMENTS

Infection fatality ratio

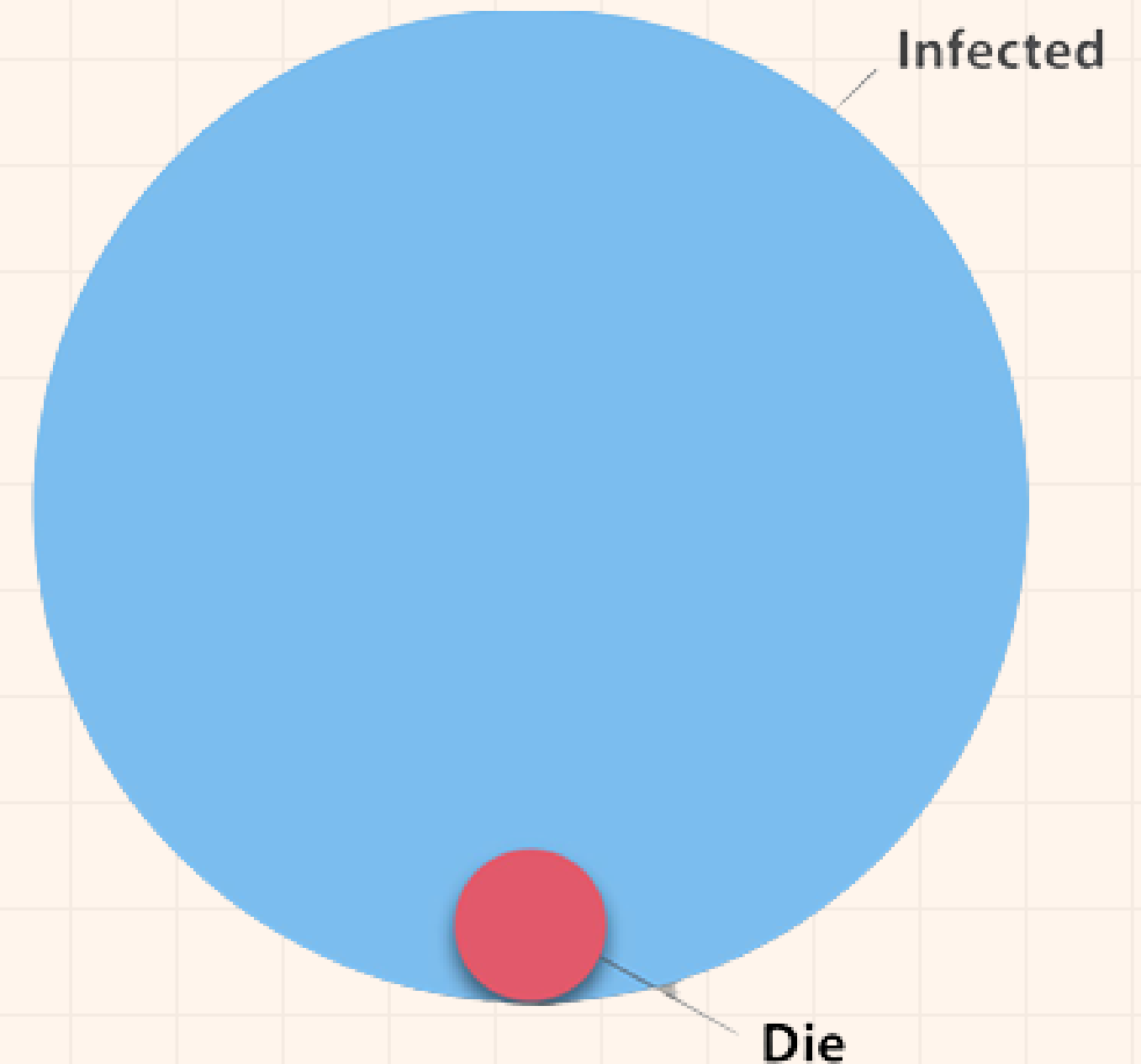
$$\frac{\text{Number who died}}{\text{Number of infected}}$$

Infection fatality risk

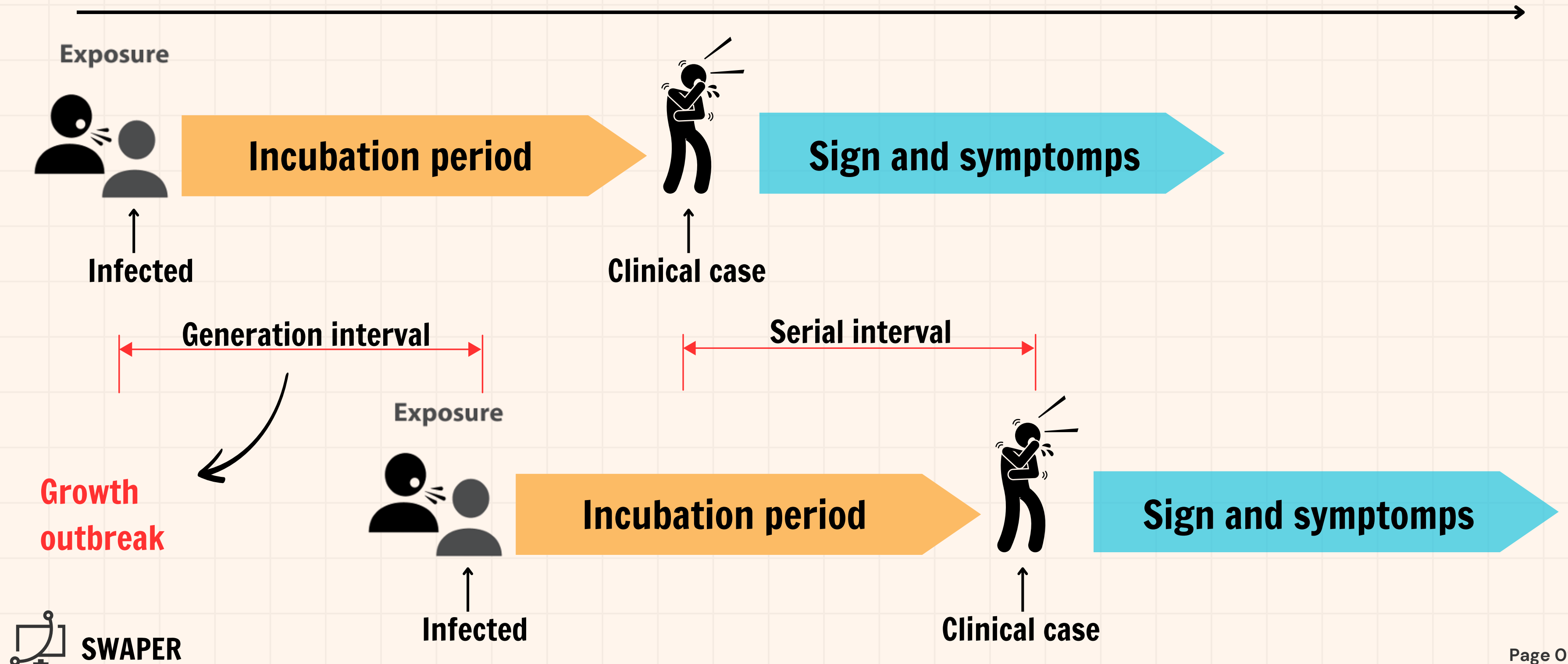
$$\frac{\text{Number who died}}{\text{Number of infected}}$$

Infection fatality rate

$$\frac{\text{Number who died}}{\text{Number of infected}}$$

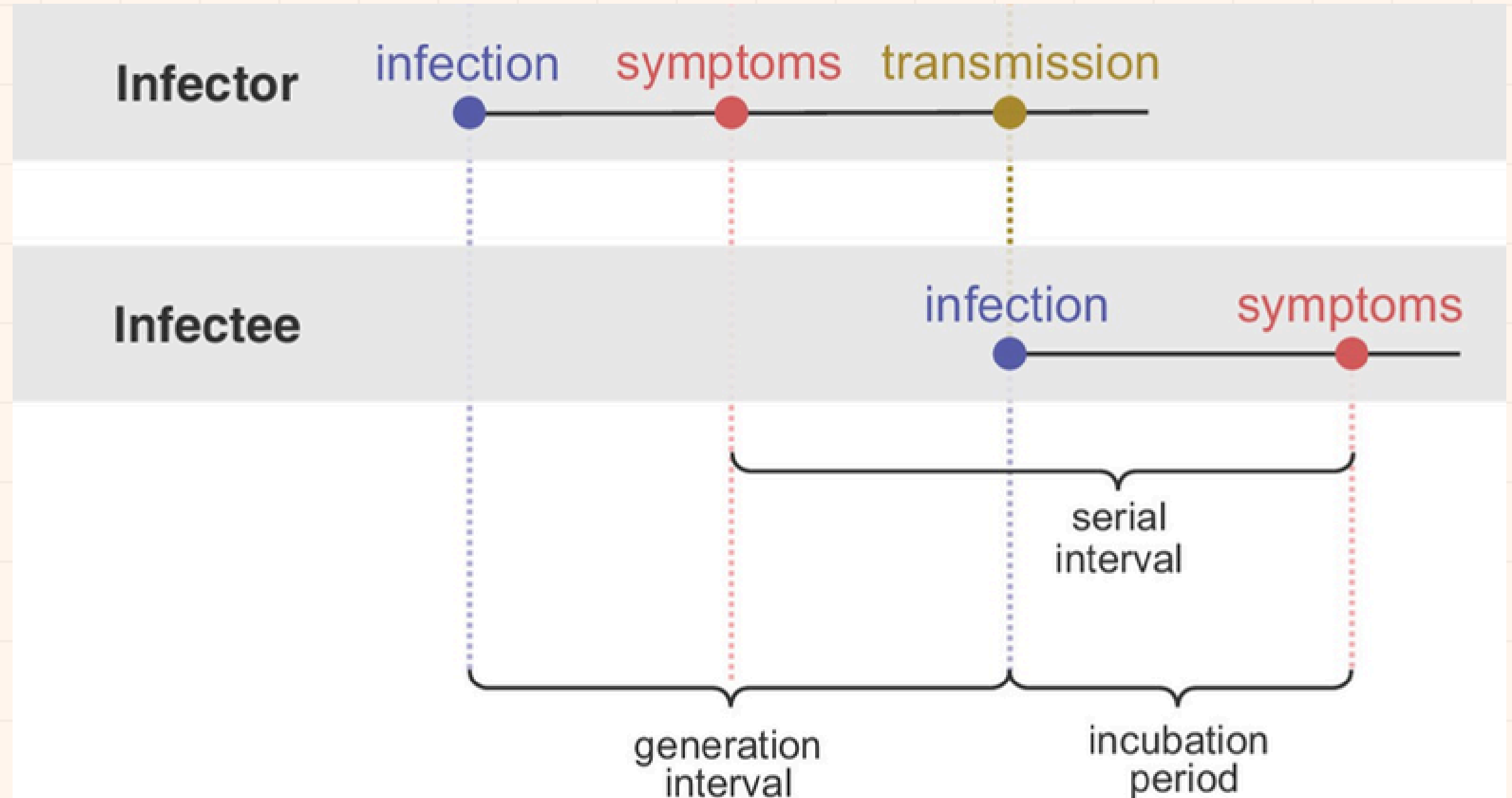


INFECTIOUS RELATIONSHIP



INFECTIOUS RELATIONSHIP

- **Serial interval can be negative** (when the symptom onset of infector is later than infectee), **generation interval is always positive.** (Gostic et al., 2020)
- The distributions of the serial interval and the generation interval **are identical in these scenarios.** (Cori et al., 2013)

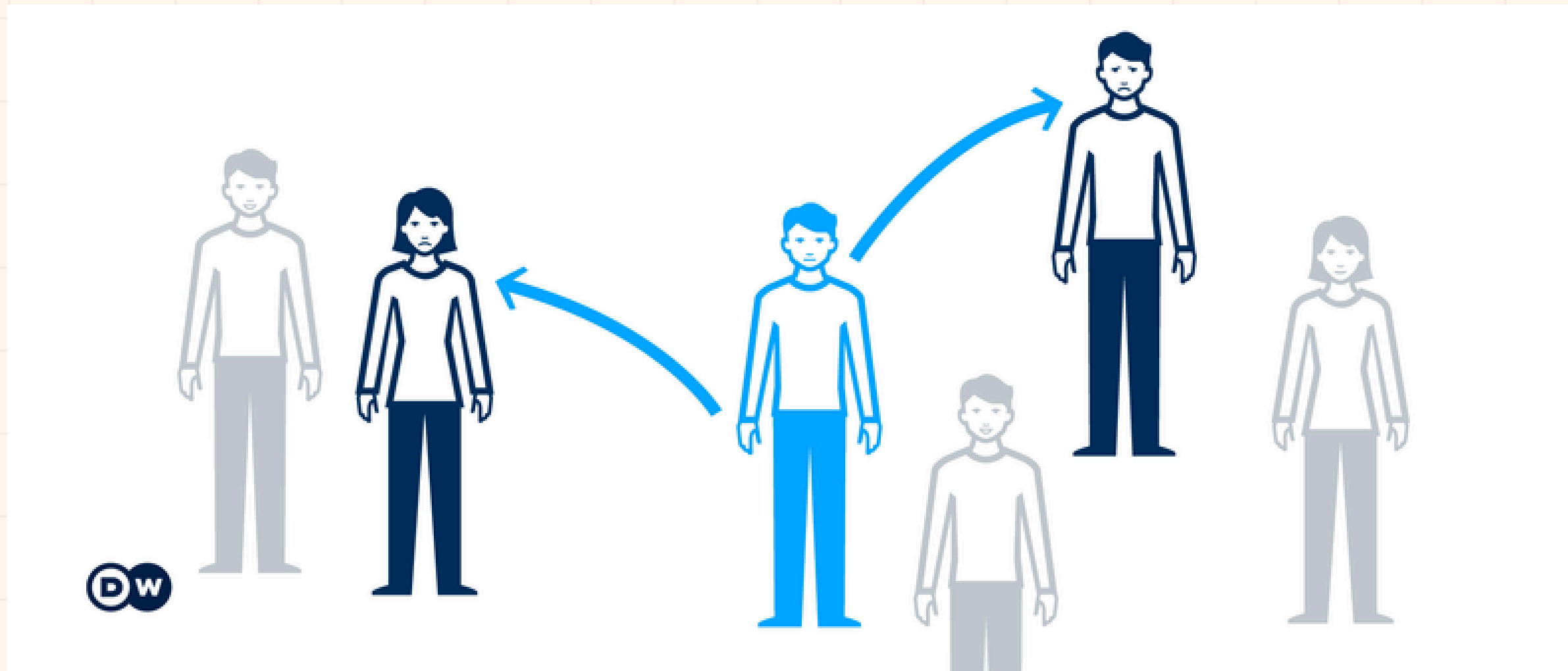


REPRODUCTIVE NUMBER

- Concept related to transmissibility
- The average number of people infected by each case
- If the reproductive number is <1 , then transmission will stop

Basic reproductive number

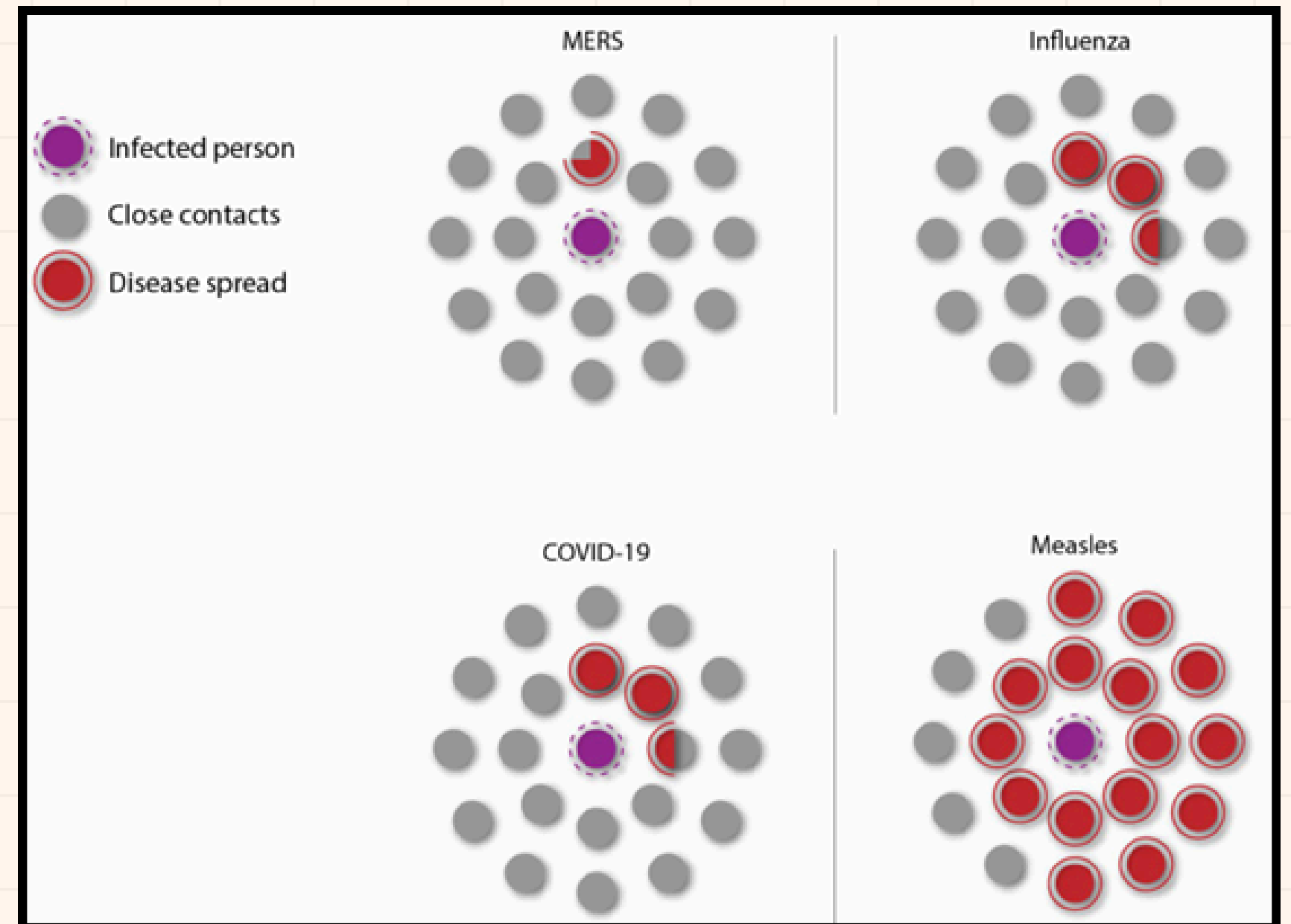
Effective reproductive number



REPRODUCTIVE NUMBER

Basic reproductive number (R_0)

- Average number of people infected by each case in a completely susceptible population
- Measure of transmissibility



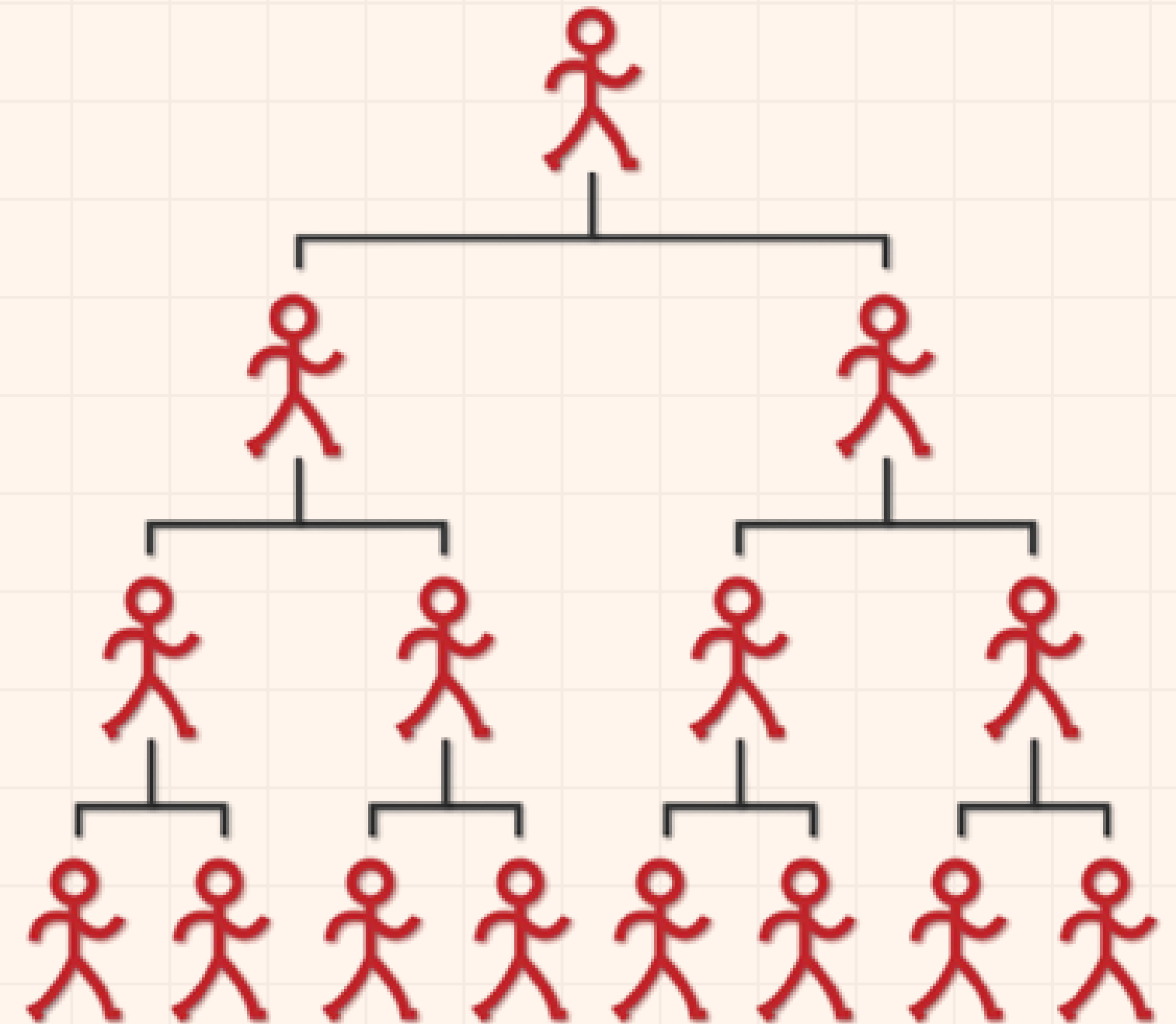
REPRODUCTIVE NUMBER

Basic reproductive number (R_0)

If each infected person
infects just two people

Size of the outbreak
doubles in each generation

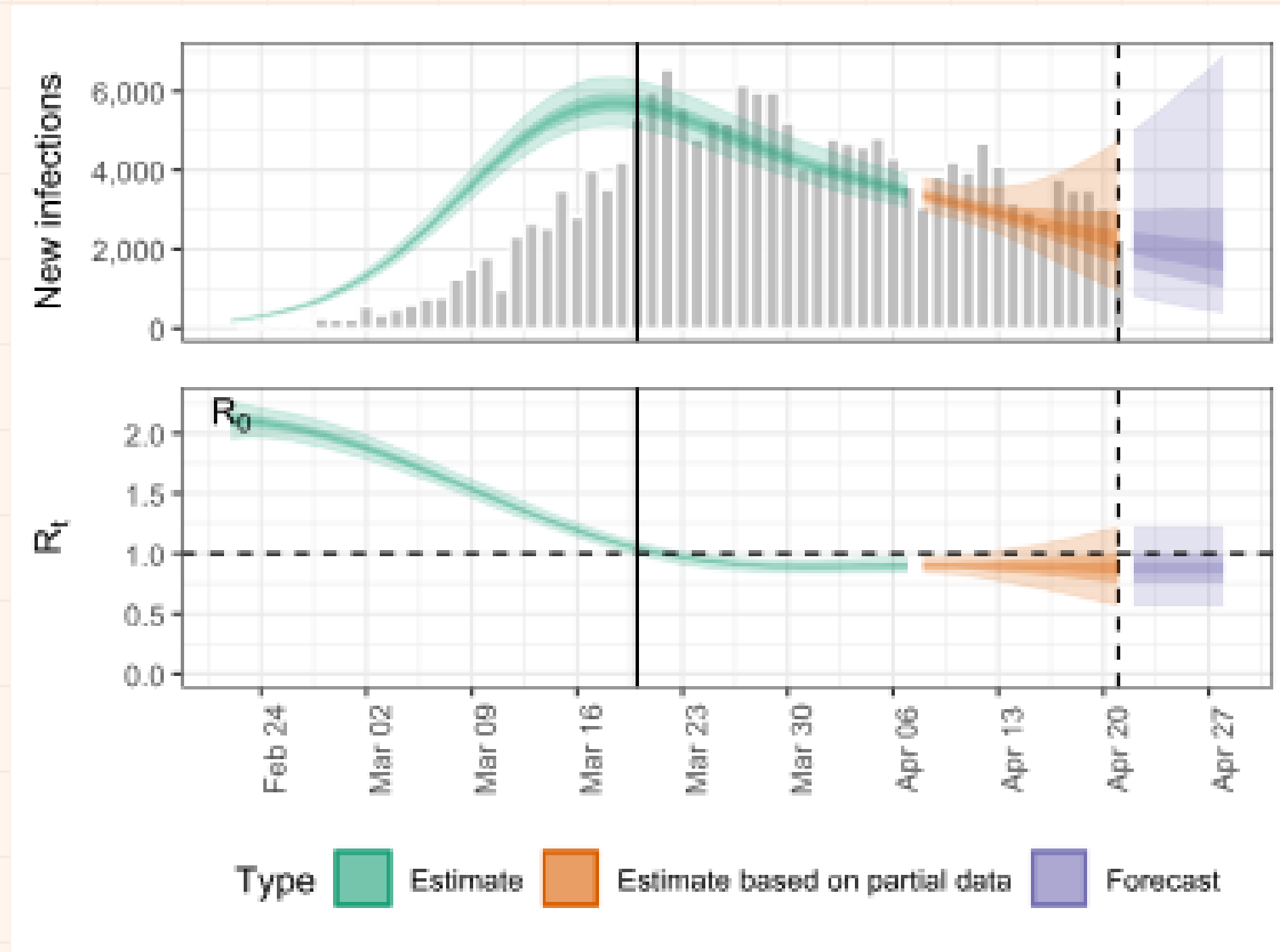
Determines Potential
Outbreak Size



REPRODUCTIVE NUMBER

Effective reproductive number (R_t)

- Average number of people infected by each case in a given place and time.
- Influenced by: immunity, interventions, other biological and behavioral factors.



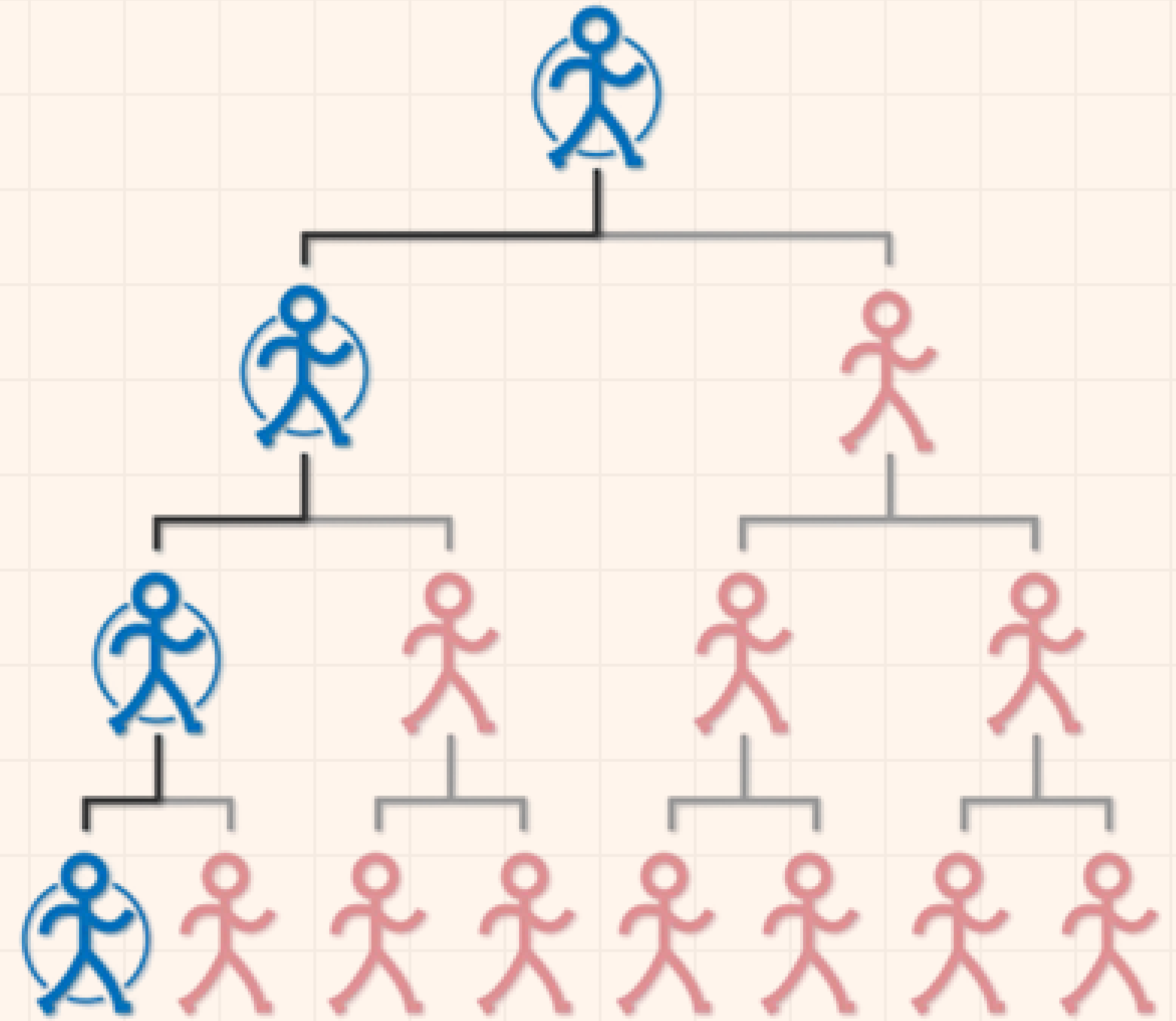
REPRODUCTIVE NUMBER

Effective reproductive number (R_t)

If we stop each case from infecting just one person

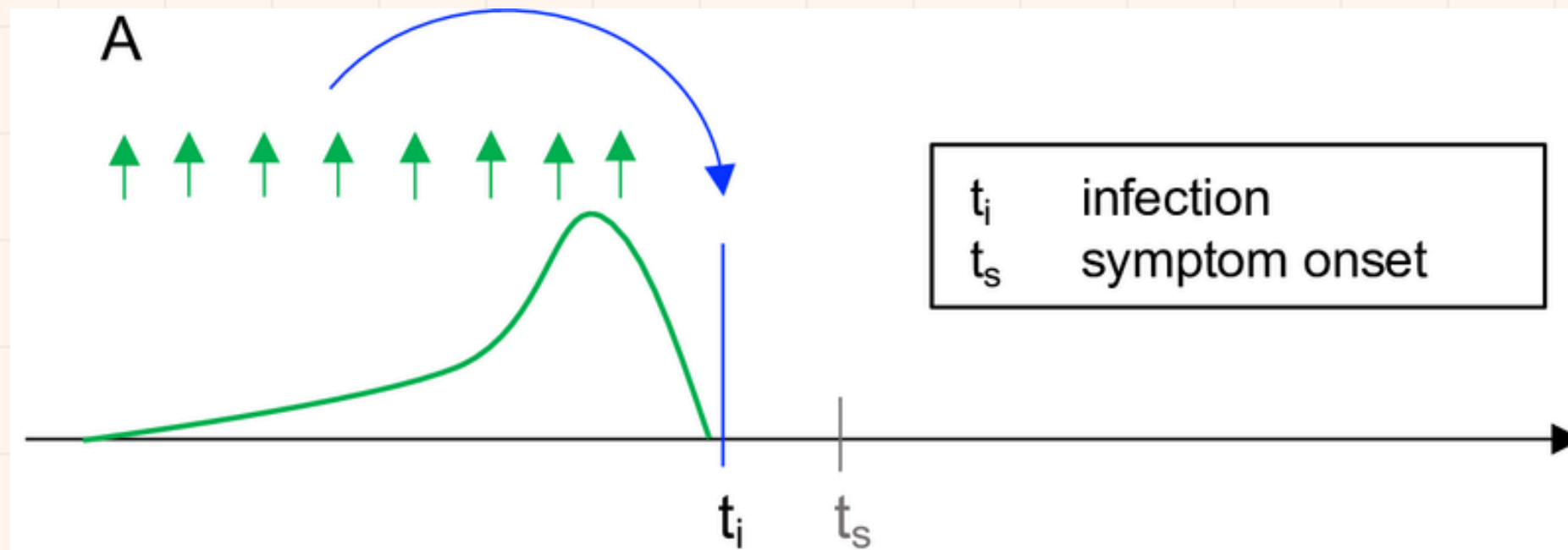
Size of the outbreak decreases

Can lead to big reductions of cases over time



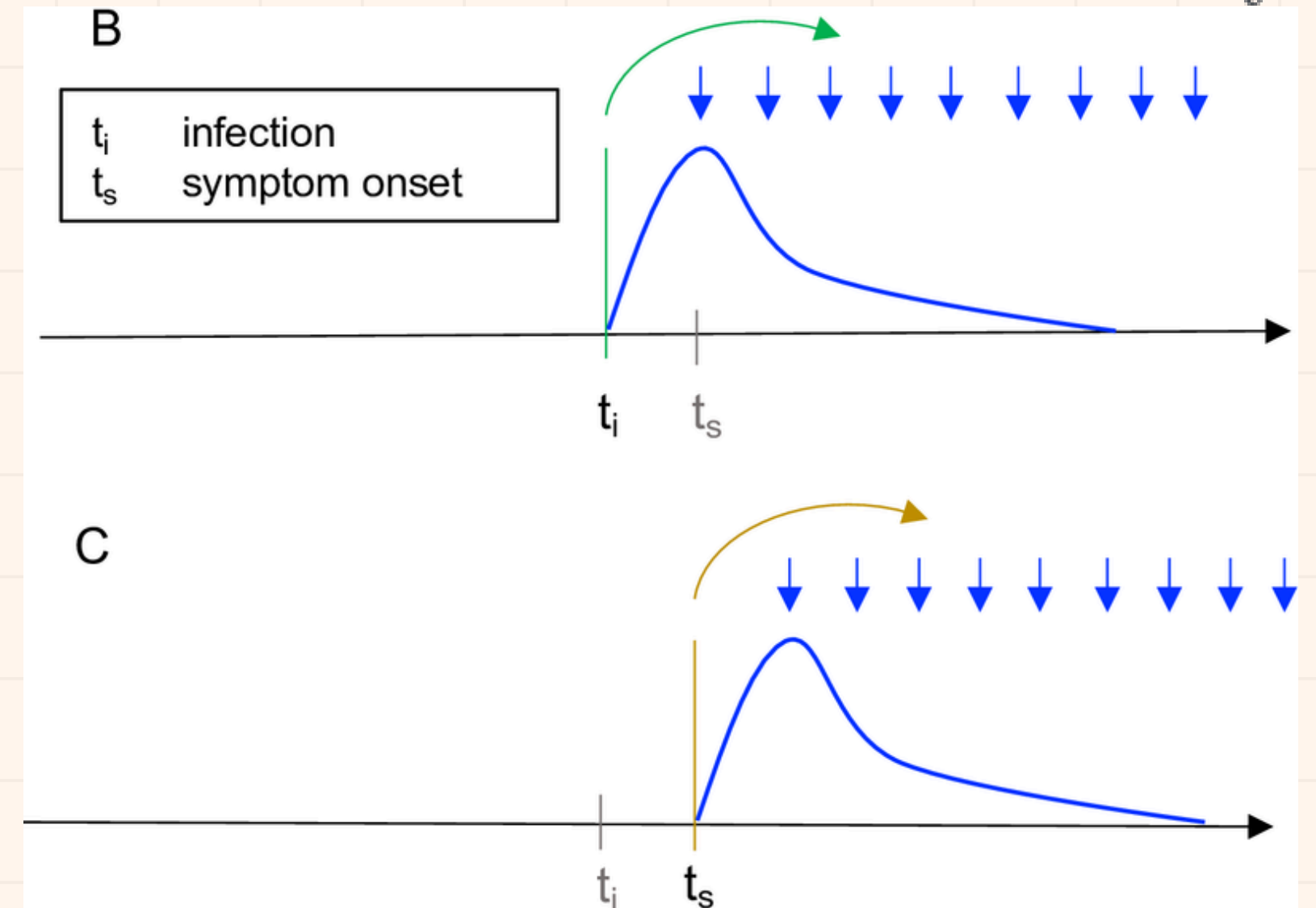
REPRODUCTIVE NUMBER

Instantaneous reproduction number R_t^i



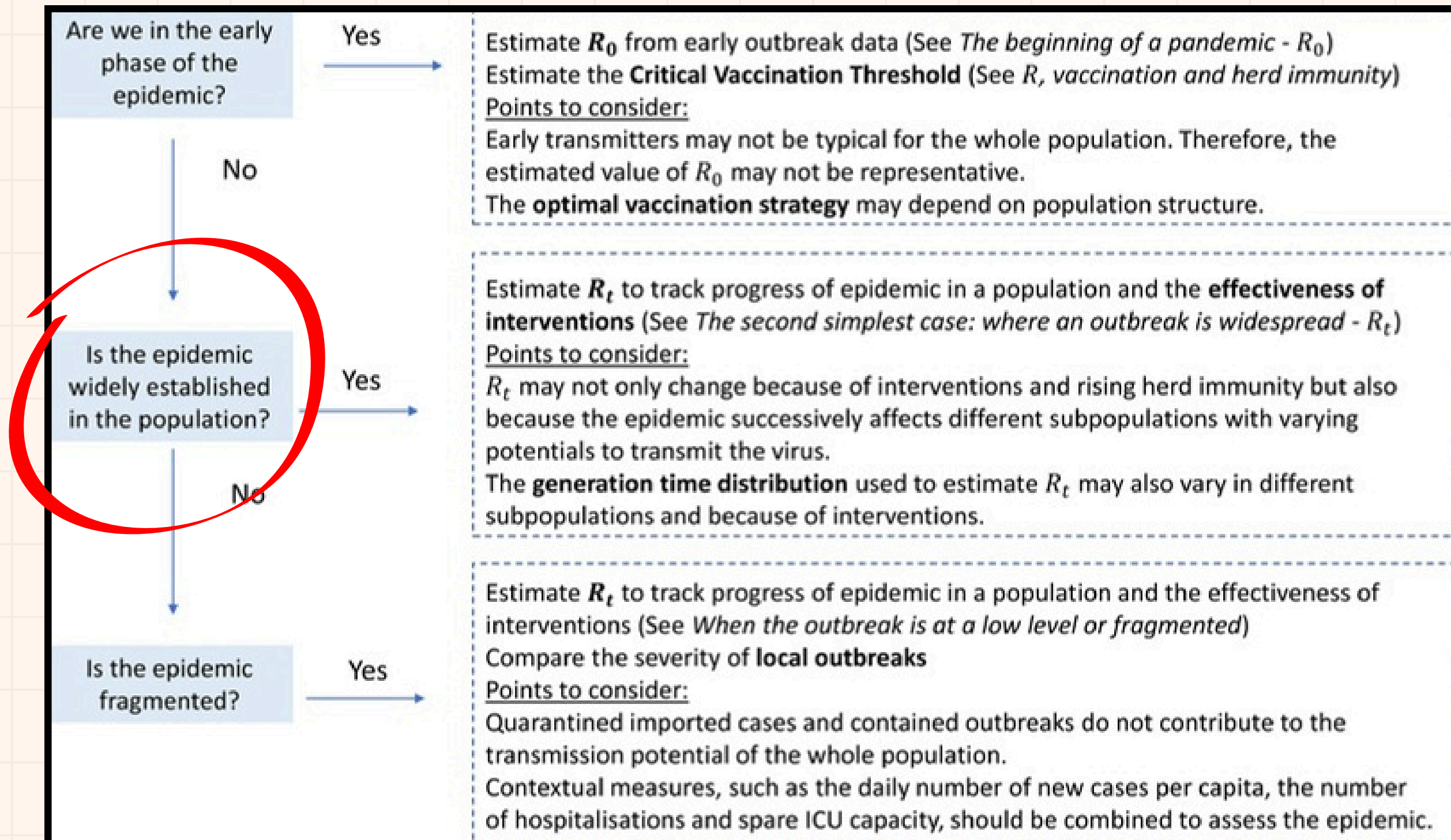
The number of new infections at a single point in time t_i (blue arrow), using the number of infections in the previous generation (green arrows) and their current infectiousness (green curve).

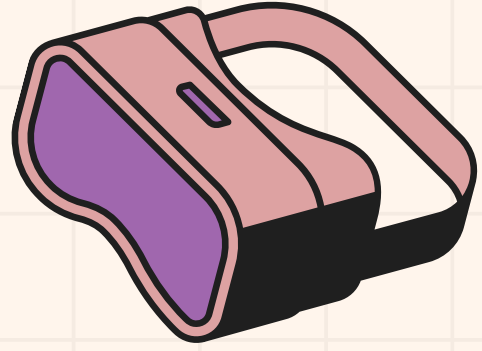
Case reproduction number R_t^c



The average number of new infections that an individual who becomes infected on day t_i (green arrows in B) or symptomatic on day t_s (yellow arrows in C) will eventually go on to cause (blue downward arrows in B and C).

REPRODUCTIVE NUMBER



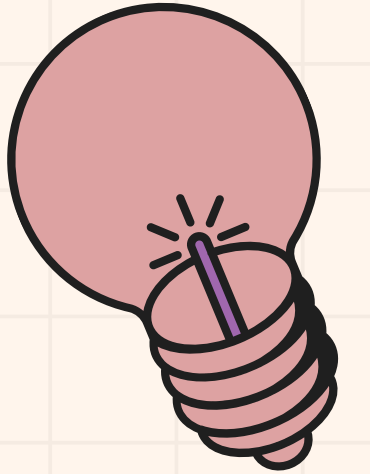


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